G-CEZO Piper PA-28-161 Cadet

Checklists and Emergency Procedures

Transcribed from the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual, Report VB-1360

Serial No. 2841226 Registration G-CEZO

This document reproduces the checklist sections of the POH word for word. The POH remains the authoritative source. Paragraph numbers in brackets refer to the amplified procedures in the POH.

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Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

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Emergency Procedures (POH Section 3)

3.2 AIRSPEEDS FOR SAFE OPERATIONS

3.2a STALL SPEEDS

2325 lbs (0° Flap) ..... 50 KIAS
2325 lbs (Full Flap ..... 44 KIAS

3.2b MANEUVERING SPEEDS

2325 lbs ..... 111 KIAS
1531 lbs ..... 88 KIAS

3.2c NEVER EXCEED SPEED

Never Exceed Speed ..... 160 KIAS

3.2d POWER OFF GLIDE SPEED

2325 lbs (0° Flap) ..... 73 KIAS

3.3 EMERGENCY PROCEDURES CHECKLIST

3.3a ENGINE FIRE DURING START (3.7)

Starter ..... CRANK ENGINE
Mixture ..... IDLE CUT-OFF
Throttle ..... OPEN
Electric Fuel Pump ..... OFF
Fuel Selector ..... OFF
Abandon if fire continues

3.3b ENGINE POWER LOSS DURING TAKEOFF (3.9)

If sufficient runway remains for a normal landing, land straight ahead.

If insufficient runway remains:
Maintain safe airspeed
Make only shallow turn to avoid obstructions
Flaps as situation requires

If sufficient altitude has been gained to attempt a restart:
Maintain safe airspeed
Fuel Selector ..... SWITCH to tank containing fuel
Electric Fuel Pump ..... CHECK ON
Mixture ..... CHECK RICH
Carburetor Heat ..... ON
Primer ..... LOCKED
If power is not regained, proceed with power off landing (3.3d).

3.3c ENGINE POWER LOSS IN FLIGHT (3.11)

Fuel Selector ..... SWITCH to tank containing fuel
Electric Fuel Pump ..... ON
Mixture ..... RICH
Carburetor Heat ..... ON
Engine Gauges ..... CHECK for indication of cause of power loss
Primer ..... CHECK LOCKED
If no fuel pressure is indicated, check tank selector position to be sure it is on a tank containing fuel.

When power is restored:
Carburetor Heat ..... OFF
Electric Fuel Pump ..... OFF

If power is not restored, prepare for power off landing (3.3d).
Trim for 73 KIAS

3.3d POWER OFF LANDING (3.13)

Locate suitable field.
Establish spiral pattern.
1000 ft. above field at downwind position for normal landing approach. When field can easily be reached slow to 63 KIAS for shortest landing.

Touchdowns should normally be made at lowest possible airspeed with full flaps extended.

When committed to landing:
Ignition ..... OFF
BATT MASTR Switch ..... OFF
ALTR Switch ..... OFF
Fuel Selector ..... OFF
Mixture ..... IDLE CUT-OFF
Seat Belts and Harnesses ..... TIGHT

3.3e FIRE IN FLIGHT (3.15)

Source of Fire ..... CHECK

Electrical fire (smoke in cabin):

BATT MASTR Switch ..... OFF
ALTR Switch ..... OFF
Vents ..... OPEN
Cabin Heat ..... OFF
Land as soon as practical.

Engine fire:

Fuel Selector ..... OFF
Throttle ..... CLOSED
Mixture ..... IDLE CUT-OFF
Electric Fuel Pump ..... CHECK OFF
Heater ..... OFF
Defroster ..... OFF
Proceed with POWER OFF LANDING procedure (3.3d).

NOTE

The possibility of an engine fire in flight is extremely remote. The procedure given is general and pilot judgement should be the determining factor for action in such an emergency.

3.3f LOSS OF OIL PRESSURE (3.17)

Land as soon as possible and investigate cause.
Prepare for power off landing (3.3d).

3.3g LOSS OF FUEL PRESSURE (3.19)

Electric Fuel Pump ..... ON
Fuel Selector ..... CHECK on TANK CONTAINING FUEL

3.3h HIGH OIL TEMPERATURE (3.21)

Land at nearest airport and investigate the problem.
Prepare for power off landing (3.3d).

3.3i ELECTRICAL FAILURES (3.23)

NOTE

When operating with light electrical load and a fully charged battery, the Alternator Inop. light may illuminate due to minimal alternator output. If the alternator is functional a slight increase in electrical load should extinguish the Inop. indication.

ALT annunciator light illuminated:

Ammeter ..... CHECK to VERIFY inop. alt.

If ammeter shows zero:
ALTR switch ..... OFF

Reduce electrical loads to minimum:
ALTNTR. FIELD Circuit Breaker ..... CHECK and RESET as required
ALTR Switch ..... ON

If power not restored:
ALTR Switch ..... OFF
If alternator output cannot be restored, reduce electrical loads and land as soon as practical. The battery is the only remaining source of electrical power.

3.3j ELECTRICAL OVERLOAD (Alternator over 20 amps above known electrical load) (3.25)

ALTR Switch ..... ON
BATT MASTR Switch ..... OFF

If alternator loads are reduced:
Electrical Load ..... REDUCE to Minimum
Land as soon as practical.

NOTE

Due to increased system voltage and radio frequency noise, operation with ALT switch ON and BATT switch OFF should be made only when required by an electrical system failure.

If alternator loads are not reduced:
ALTR switch ..... OFF
BATT MASTR Switch ..... AS REQUIRED
Land as soon as possible. Anticipate complete electrical failure.

3.3k SPIN RECOVERY (3.27)

Throttle ..... IDLE
Ailerons ..... NEUTRAL
Rudder ..... FULL OPPOSITE to DIRECTION of ROTATION
Control Wheel ..... FULL FORWARD
Rudder ..... NEUTRAL (when rotation stops)
Control Wheel . AS REQUIRED to SMOOTHLY REGAIN LEVEL FLIGHT ATTITUDE

3.3m OPEN DOOR (3.29)

If both upper and lower latches are open, the door will trail slightly open and airspeeds will be reduced slightly.

To close the door in flight:
Slow airplane to 89 KIAS
Cabin Vents ..... CLOSE
Storm Window ..... OPEN
If upper latch is open ..... LATCH
If side latch if open PULL on ARMREST while moving latch handle to LATCH position.
If both latches are open ..... LATCH SIDE LATCH then TOP LATCH.

3.3n ENGINE ROUGHNESS (3.31)

Carburetor Heat ..... ON

If roughness continues after one minute:
Carburetor Heat ..... OFF
Mixture ..... ADJUST for MAXIMUM SMOOTHNESS
Electric Fuel Pump ..... ON
Fuel Selector ..... SWITCH TANKS
Engine Gauges ..... CHECK
Magneto Switch ..... L then R then BOTH

If operation is satisfactory on either magneto, continue on that magneto at reduced power and full RICH mixture to first airport.

Prepare for power off landing (3.3d).

NOTE

Partial carburetor heat may be worse than no heat at all, since it may melt part of the ice which will refreeze in the intake system. Therefore, when using carburetor heat always use full heat; and, when ice is removed, return the control to the full cold position.

3.3o CARBURETOR ICING (3.33)

Carburetor Heat ..... ON
Mixture ..... ADJUST for MAXIMUM SMOOTHNESS

Normal Procedures (POH Section 4)

4.3 AIRSPEEDS FOR SAFE OPERATIONS

(a) Best Rate of Climb Speed ..... 79 KIAS
(b) Best Angle of Climb Speed ..... 63 KIAS
(c) Turbulent Air Operating Speed (See Subsection 2.3) ..... 111 KIAS
(d) Maximum Flap Speed ..... 103 KIAS
(e) Landing Final Approach Speed (Flaps 40°) ..... 63 KIAS
(f) Maximum Demonstrated Crosswind Velocity ..... 17 KTS

4.5 NORMAL PROCEDURES CHECKLIST

4.5a Preparation (4.9)

Airplane Status ..... AIRWORTHY, PAPERS ON BOARD
Weather ..... SUITABLE
Baggage ..... WEIGHED, STOWED, TIED
Weight and C.G. ..... WITHIN LIMITS
Navigation ..... PLANNED
Charts and Navigation Equipment ..... ON BOARD
Performance and Range ..... COMPUTED AND SAFE

CAUTION

The flap position should be noted before boarding the airplane. The flaps must be placed in the UP position before they will lock and support weight on the step.

4.5b Preflight Checklists (4.11)

COCKPIT (4.11a)

Control Wheel ..... RELEASE BELTS
Circuit Breakers ..... IN
Avionics ..... OFF
Parking Brake ..... SET
Electric Switches ..... OFF
Magneto Switch ..... OFF
Mixture ..... IDLE CUT-OFF
BATT MASTR Switch ..... ON
Fuel Quantity Gauges ..... CHECK
Annunciator Panel ..... CHECK
BATT MASTR Switch ..... OFF
Flight Controls ..... CHECK
Flaps ..... CHECK

Trim ..... CHECK, SET NEUTRAL
Pitot Drain ..... DRAIN, CLOSE
Static Drain ..... DRAIN, CLOSE
Windows ..... CHECK, CLEAN
Tow Bar ..... STOW
Baggage ..... SECURE

RIGHT WING (4.11b)

Wing ..... FREE of ICE, SNOW, FROST
Control Surfaces ..... CHECK for interference - CLEAR of ICE, SNOW, FROST
Hinges ..... CHECK for interference
Static Wicks ..... CHECK
Wing Tip and Lights ..... CHECK
Fuel Tank ..... CHECK supply visually - SECURE CAPS

CAUTION

When draining any amount of fuel, care should be taken to ensure that no fire hazard exists before starting engine.

Fuel Tank Sump ..... DRAIN, CHECK for water, sediment and proper fuel
Fuel Vent ..... CLEAR
Tie Down and Chock ..... REMOVE
Main Gear Strut ..... PROPER INFLATION (4.50 in.)
Tire ..... CHECK
Brake Block and Disc ..... CHECK
Fresh Air Inlet ..... CLEAR

NOSE SECTION (4.11c)

Fuel and Oil ..... CHECK FOR LEAKS
Cowling ..... SECURE
Windshield ..... CLEAN
Propeller and Spinner ..... CHECK
Air Inlets ..... CLEAR
Alternator Belt ..... CHECK TENSION
Landing Light ..... CHECK
Nose Chock ..... REMOVE
Nose Gear Strut ..... PROPER INFLATION (3.25 in.)
Nose Wheel Tire ..... CHECK
Oil ..... CHECK LEVEL
Dipstick ..... PROPERLY SEATED
Fuel Strainer ..... DRAIN, CHECK for water, sediment and proper fuel

LEFT WING (4.11d)

Wing ..... FREE of ICE, SNOW, FROST
Fresh Air Inlet ..... CLEAR

Main Gear Strut ..... PROPER INFLATION (4.50 in.)
Tire ..... CHECK
Brake Block and Disc ..... CHECK
Fuel Tank Sumps ..... DRAIN, CHECK for water, sediment and proper fuel
Fuel Vent ..... CLEAR
Tie Down and Chock ..... REMOVE
Fuel Tanks ..... CHECK supply visually - SECURE CAPS
Pitot/Static Head ..... REMOVE COVER - HOLES CLEAR
Wing Tip and Lights ..... CHECK
Control Surfaces ..... CHECK for interference - FREE of ICE, SNOW, FROST
Hinges ..... CHECK for interference
Static Wicks ..... CHECK

FUSELAGE (4.11e)

Antennas ..... CHECK
Empennage ..... FREE of ICE, SNOW, FROST
Fresh Air Inlet ..... CLEAR
Stabilator and Trim Tab ..... CHECK for interference
Tie Down ..... REMOVE
BATT MASTR Switch ..... ON
Cockpit Lighting ..... CHECK
Nav and Strobe Lights ..... CHECK
Stall Warning ..... CHECK
Pitot Heat ..... CHECK
All Switches ..... OFF
Passengers ..... BOARD
Cabin Door ..... CLOSE and SECURE
Seat Belts and Harnesses ..... FASTEN - CHECK interia reel

4.5c Before Starting Engine Checklist (4.13)

BEFORE STARTING ENGINE (4.13)

Brakes ..... SET
Carburetor Heat ..... FULL OFF
Fuel Selector ..... DESIRED TANK
Avionics ..... OFF
ALTR Switch ..... ON

4.5d Engine Start Checklist (4.15)

NORMAL START - COLD ENGINE (4.15a)

Throttle ..... ¼ INCH OPEN
BATT MASTR Switch ..... ON
Electric Fuel Pump ..... ON
Mixture ..... FULL RICH
Starter ..... ENGAGE
Throttle ..... ADJUST
Oil Pressure ..... CHECK
If engine does not start within 10 sec., prime and repeat starting procedure.

NORMAL START - HOT ENGINE (4.15b)

Throttle ..... ½ INCH OPEN
BATT MASTR Switch ..... ON
Electric Fuel Pump ..... ON
Mixture ..... FULL RICH
Starter ..... ENGAGE
Throttle ..... ADJUST
Oil Pressure ..... CHECK

ENGINE START WHEN FLOODED (4.15c)

Throttle ..... OPEN FULL
BATT MASTR Switch ..... ON
Electric Fuel Pump ..... OFF
Mixture ..... IDLE CUT-OFF
Starter ..... ENGAGE
Mixture ..... ADVANCE
Throttle ..... RETARD
Oil Pressure ..... CHECK

ENGINE START WITH EXTERNAL POWER SOURCE (4.15d)

BATT MASTR Switch ..... OFF
All Electrical Equipment ..... OFF
Terminals ..... CONNECT
External Power Plug ..... INSERT in receptacle

NOTE

For all normal operations using the PEP jumper cables, the battery master switch should be OFF. However, it is possible to use the ship's battery in parallel by turning the battery master switch ON. This will give longer cranking capabilities, but will not increase the amperage.

CAUTION

Care should be exercised, because, if the ship's battery has been depleted, the external power supply can be reduced to the level of the ship's battery. This can be tested by turning the battery master switch ON momentarily while the starter is engaged. If cranking speed increases, the ship's battery is at a higher level than the external power supply.

Proceed with normal start
Throttle ..... LOWEST POSSIBLE RPM
External Power Plug ..... DISCONNECT from receptacle
BATT MASTR Switch ..... ON - CHECK AMMETER
Oil Pressure ..... CHECK

4.5e Engine Warm-Up Checklist (4.17)

ENGINE WARM-UP (4.17)

Throttle ..... 800 to 1200 RPM

4.5f Taxiing Checklist (4.19)

TAXIING (4.19)

Parking Brake ..... RELEASE
Taxi Area ..... CLEAR
Throttle ..... APPLY SLOWLY
Brakes ..... CHECK
Steering ..... CHECK

4.5g Ground Check Checklist (4.21)

GROUND CHECK (4.21)

Parking Brake ..... SET
Throttle ..... 2000 RPM
Magnetos ..... max. drop 175 RPM -max. diff. 50 RPM
Vacuum ..... 4.8 to 5.1 in. Hg
Oil Temperature ..... CHECK
Oil Pressure ..... CHECK
Air Conditioner ..... CHECK
Annunciator Panel ..... PRESS-TO-TEST
Carburetor Heat ..... CHECK
Engine is warm for takeoff when throttle can be opened without engine faltering.
Electric Fuel Pump ..... OFF
Fuel Pressure ..... CHECK
Throttle ..... RETARD

4.5h Before Takeoff Checklist (4.23)

BEFORE TAKEOFF (4.23)

Battery Master Switch ..... ON
Alternator Switch ..... ON
Flight Instruments ..... CHECK
Fuel Selector ..... PROPER TANK
Electric Fuel Pump ..... ON
Engine Gauges ..... CHECK
Carburetor Heat ..... OFF
Seat Backs ..... ERECT

NOTE

The mixture should be set FULL RICH, but a minimum amount of leaning is permitted for smooth engine operation when taking off at high elevation.

Mixture ..... SET
Primer ..... LOCK

NOTE

The inertia reel type shoulder harness should be given a pull test to check its locking restraint feature.

NOTE

If the fixed shoulder harness (non-inertial reel type) are installed for the aft seat occupants they should be adjusted to provide adequate restraint.

Belts Harness ..... FASTENED CHECK
Empty Seats ..... SEAT BELTS SNUGLY FASTENED
Flaps ..... SET
Trim ..... SET
Controls ..... FREE
Door ..... LATCHED
Air Conditioner ..... OFF

4.5i Takeoff Checklist (4.25)

NORMAL (4.25a)

Flaps ..... SET
Trim ..... SET
Accelerate to 45 to 55 KIAS
Control Wheel ..... back pressure to rotate to climb attitude

0° FLAPS TAKEOFF PERFORMANCE (4.25b)

Flaps ..... UP
Accelerate to 40-50 KIAS (depending on weight)
Control Wheel ..... back pressure to rotate to climb attitude

Accelerate to and maintain 44 to 55 KIAS (depending on weight) until obstacle clearance is achieved and climb out at 79 KIAS.

25° FLAPS TAKEOFF PERFORMANCE (4.25c)

Flaps ..... 25° (second notch)
Accelerate to 40-50 KIAS (depending on weight)
Control Wheel ..... back pressure to rotate to climb attitude

Accelerate to and maintain 44 to 55 KIAS (depending on weight) until obstacle clearance is achieved and climb out at 79 KIAS.
Flaps ..... RETRACT SLOWLY

SOFT FIELD, OBSTACLE CLEARANCE (4.25d)

Flaps ..... 25° (second notch)

Accelerate and lift off nose gear as soon as possible. Lift off at lowest possible airspeed. Accelerate just above ground to 52 KIAS to climb past obstacle height. Continue climbing while accelerating to best rate of climb speed, 79 KIAS.
Flaps ..... RETRACT SLOWLY

SOFT FIELD, NO OBSTACLE (4.25e)

Flaps ..... 25° (second notch)

Accelerate and lift off nose gear as soon as possible. Lift off at lowest possible airspeed. Accelerate just above ground to best rate of climb speed, 79 KIAS.
Flaps ..... RETRACT SLOWLY

4.5j Climb Checklist (4.27)

CLIMB (4.27)

Best Rate Climb Speed (Flaps Up) ..... 79 KIAS
Best Angle Climb Speed (Flaps Up) ..... 63 KIAS
En Route ..... 87 KIAS
Electric Fuel Pump ..... OFF at desired altitude

4.5k Cruise Checklist (4.29)

CRUISE

Reference performance charts and Avco-Lycoming Operators Manual.
Normal Maximum Power 75%
Power ..... SET per power table
Mixture ..... ADJUST

4.5m Descent Checklist (4.31)

NORMAL DESCENT (4.31a)

Throttle ..... 2500 rpm
Airspeed ..... 126 KIAS
Mixture ..... RICH
Carburetor Heat ..... ON if required

POWER OFF DESCENT (4.31b)

Carburetor Heat ..... ON if required
Throttle ..... CLOSE
Airspeed ..... AS REQUIRED
Mixture ..... AS REQUIRED
Power ..... VERIFY with throttle EVERY 30 SECONDS

4.5n Approach And Landing Checklist (4.33)

APPROACH AND LANDING (See charts in Section 5) (4.33)

Fuel Selector ..... PROPER TANK
Seat Backs ..... ERECT
Belts/Harness ..... FASTEN/CHECK

NOTE

If the fixed shoulder harness (non-inertia reel type) is installed, it must be connected to the seat belt and adjusted to allow proper accessibility to all controls including fuel selector, flaps, trim, etc, while maintaining adequate restraint for the occupant.

NOTE

If the inertia reel type shoulder harness is installed, a pull test of its locking restraint feature should be performed.

Electric Fuel Pump ..... ON
Mixture ..... SET
Flaps ..... SET - 103 KIAS MAXIMUM
Air Conditioner ..... OFF
Trim to 70 KIAS
Final Approach Speed (Flaps 40°) ..... 63 KIAS

4.5o Stopping Engine Checklist (4.35)

STOPPING ENGINE (4.35)

Flaps ..... RETRACT
Electric Fuel Pump ..... OFF
Electrical Switches ..... OFF
Air Conditioner ..... OFF
Radios ..... OFF
Throttle ..... FULL AFT
Mixture ..... IDLE CUT-OFF
Magnetos ..... OFF
ALTR Switch ..... OFF
BATT MASTR Switch ..... OFF
Trim ..... SET

NOTE

When alternate fuels are used, the engine should be run up to 1200 rpm for one minute prior to shutdown to clean out any unburned fuel.

NOTE

The flaps must be placed in the UP position for the flap step to support weight. Passengers should be cautioned accordingly.

4.5p Mooring Checklist (4.37)

MOORING (4.37)

Parking Brake ..... SET
Control Wheel ..... SECURE with belts
Flaps ..... FULL UP
Wheel Chocks ..... IN PLACE
Tie Downs ..... SECURE

Cockpit Placards (POH 2.23)

TAKEOFF CHECKLIST

In full view of the pilot:

Fuel on proper tank
Electric fuel pump on
Engine gauges checked
Flaps - set
Carb. heat off
Mixture set
Primer locked
Seat backs erect
Fasten belts/harness
Trim tab - set
Controls - free
Door - latched
Air conditioner off

LANDING CHECKLIST

Fuel on proper tank
Mixture rich
Electric fuel pump on
Seat backs erect
Flaps - set (White Arc)
Fasten belts/harness
Air conditioner off

In air conditioned aircraft, the AIR COND OFF item in the takeoff and landing checklists above, is mandatory.